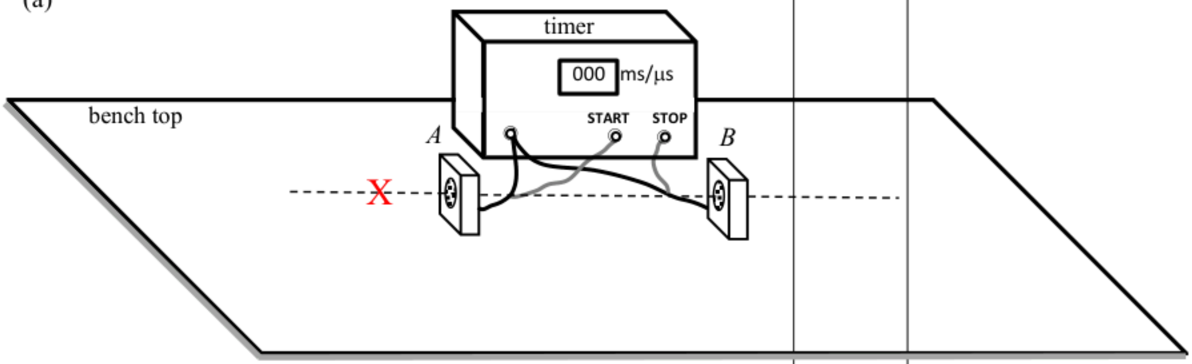
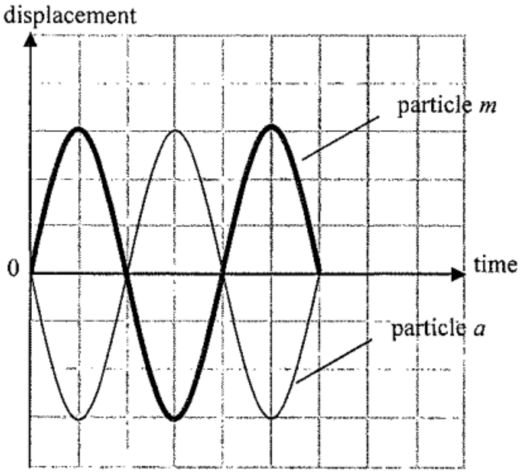
6. (a) 		
Correct location of 'X' (close to axis and on the side of the START microphone).	1A	
Use <u>a metre rule</u> to measure the distance / separation D between the two microphones.	1A	Accept ruler
Record the timer reading, which is the time interval Δt for the sound to travel from one microphone (START) to the other microphone (STOP), i.e. A to B.	1A	Any ONE of the measurements
	3	
(b) (i) Discard 539 µs, $\Delta t = \dfrac{801+838+821}{3} = 820\ \mu s$ The speed of sound in air $v = \dfrac{D}{\Delta t}$ $v = \dfrac{0.280}{820 \times 10^{-6}}$	1M	1M for taking the average value of Δt and quoting the formula $v = \frac{D}{\Delta t}$ to calculate the speed of sound, where D is the separation between the microphones.
$= 341.463415\ m\ s^{-1} \approx 341\ m\ s^{-1}$	1A	Accept 340 m s^{-1} to 342 m s^{-1}
(ii) Increase separation D between the microphones.	1A	
	3	

Solution	Marks	Remarks
2. (a) $v = \frac{30 \times 2}{0.04}$	1M	
$= 1500 \text{ m s}^{-1}$	1A	
	2	
(b) (i) $\frac{p_1}{T_1} = \frac{p_2}{T_2}$		
$\frac{18.5}{273+27} = \frac{p_2}{273+20}$	1M	
$p_2 = 18.068333 \text{ atm} \approx 18.1 \text{ atm}$		
	1	→ kinetic theory
(ii) As the temperature decreases, the speed / kinetic energy of the gas molecules decreases.	1A	
They collide less frequently and/or violently with the wall of the container / cylinder.	1A	
Thus, the pressure decreases.		
	2	
(c) (i) $n_0 = n_1 + n_2$	1M	Assume no leakage
$p_0V_0 = p_1V_1 + p_2V_2$ fixed volume (container)		
$18.1 \times 0.012 = p_1 \times 0.012 + 4.0 \times 0.015$	1M	
$p_1 = 13.1 \text{ atm}$ (i.e. $\Delta p = 5.0 \text{ atm}$)		
	2	
(ii) The pressure decreases by 5.0 atm for inflating one balloon, assume k balloons can be inflated completely:		Accept:
$18.1 - 5.0k \geq 4.0$ or $k \leq \frac{18.1-4.0}{5.0} = 2.82$	1M	18.1 atm → 13.1 atm → 8.1 atm
$\therefore k = 2$ ↳ seabed atm, 出唔到氣	1A	$\therefore k = 2$
<u>Or</u> Let V = volume of compressed gas under 4.0 atm $18.1 \times 0.012 = 4.0 \times V$ $V = 0.0543 \text{ m}^3$ $0.0543 - 0.015 \times k \geq 0.012$	1M	
$k \leq 2.82$ $\therefore k = 2$	1A	
	2	

Solution	Marks	Remarks
6. (a) Sound waves having the same frequency / wavelength and a constant phase difference (or always in phase / in opposite phase) are coherent.	1A	
	1	
(b) (i) When the sound waves from A and B meet at various positions along OY, interference occurs.		
At positions where the two waves are in phase, constructive interference occurs and gives maximum (loudness).	1A	
At positions where the two waves are in opposite phase, destructive interference occurs and gives minimum (loudness).	1A	
	2	
(ii) Any ONE: • due to background noise • due to reflection of unwanted sound from the wall, floor etc. • the intensity of sound waves from A and B reaching P are NOT equal as $AP < BP$ therefore cancellation is incomplete.	1A	
	1	
(c) Path difference at Q $3\lambda/2 = 2.58 - 2.17$ $= 0.41$ m $\therefore\ \lambda = 0.27333333$ m ≈ 0.273 m	1M	
$v = f\lambda$ $= (1200)(0.273)$ $= 328$ m s^{-1}	1A	Accept 327.6 m s^{-1} to 328 m s^{-1}
	2	
(d) Path difference Δ at any position along OY from A and B is less than AB (i.e. $\Delta = n\lambda < AB$) max. $\Delta = 0.80$ m $= \frac{0.8}{0.273}\lambda \approx 2.93\lambda$ i.e. max. $\Delta < 3\lambda$	1M	
As path difference cannot be equal to $3\lambda, 4\lambda, \cdots$, therefore no more maximum beyond R.	1A	
	2	
(e) increases	1A	
	1	

5. (a)	(i)	*a* / *i* / *q*	1A	
			1	
	(ii)	*c* / *k* / *s*	1A	
			1	
(b)	(i)	wavelength (4 grids) = 4 × 50 cm = 200 cm or 2 m	1A	
			1	
	(ii)	speed = frequency × wavelength frequency = speed ÷ wavelength = 340 ÷ 2 = 170 Hz	1M 1A	
			2	
	(iii)		2A	
			2	
(c)		Displacement increases, i.e. further to the left.	1A	
			1	

7. (a)	The speed of sound increases (linearly) with air temperature. <u>Or</u> The higher the air temperature, the greater is the speed of sound.	1A		
			1	
(b)	The air near the ground is cooler while the air layers higher above are relatively warmer, the sound wave bends downward (towards the normal) as it travels slower when going from a higher temperature (upper) layer into a low temperature (lower) layer and vice versa, i.e. refraction occurs.	1A 1A		
			2	
(c) (i)	Speed ratio $\frac{3.00\times10^8}{337} = 8.90208\times10^5 \approx 8.90\times10^5$	1A		
			1	
(ii)	The distance is given by 337×3.0 $= 1011$ m	 1M 1A		e.c.f. from (c)(i) using incorrect sound speed Accept: 1010 m ~ 1011 m
			2	